Dev Popli

d1e1v1234@gmail.com | +91-9416325144
B.Tech – Computer Science and Engineering, IIIT Una
GitHub | LinkedIn | LeetCode
CGPA: **8.88 / 10.0** (2024–Present)

Summary

Third-year B.Tech CSE student at IIIT Una with a CGPA of **8.88/10**, experienced in building and deploying end-to-end Machine Learning and Deep Learning applications. Skilled in **C++, Python, SQL, Data Structures and Algorithms, TensorFlow, and Scikit-learn**, with **250+ DSA problems** solved on LeetCode. Hands-on experience in computer vision, predictive modelling, model deployment, and real-time ML applications. Seeking Software Engineering and AI/ML internship opportunities.

Technical Skills

- **Programming Languages**: C++, Python, C, SQL, JavaScript
- **Machine Learning & Deep Learning**: Scikit-learn, TensorFlow, Keras, CNNs, Random Forest, XGBoost, Regression, Classification, Feature Engineering, Hyperparameter Tuning , Generative AI , RAG
- **Data Analysis**: NumPy, Pandas, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Data Preprocessing, Feature Selection
- **Computer Vision**: OpenCV, Image Classification, Real-Time Video Processing
- **Web & Deployment**: Streamlit, Flask, FAST APIs, Docker, HTML, CSS
- **Databases**: MySQL, Relational Databases
- **Developer Tools**: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Kaggle
- **Core Computer Science**: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems

Projects

•**Driver Drowsiness Detection | Real-Time Deep Learning System** *June 2026*
TensorFlow, Keras, MobileNetV3, OpenCV, Streamlit, WebRTC, Docker GitHub | Live Demo

- –Developed a **real-time driver drowsiness detection system** using TensorFlow and MobileNetV3-based CNN models to independently classify **eye state (open/closed)** and detect **yawning** from live webcam input.
- –Built an **OpenCV-based computer vision pipeline** for face and eye region extraction and integrated **streamlit-webrtc** for low-latency, browser-based real-time video processing and model inference.
- –Achieved **%97 test accuracy** for eye-state classification and **96% test accuracy** for yawn detection on held-out test data.
- –Containerised the application using **Docker** and deployed an interactive web application on **Streamlit Community Cloud** for real-time drowsiness monitoring.

•**Hospital LOS & Billing Prediction | Machine Learning System** *May 2026*
Python, Scikit-learn, Random Forest, Pandas, Feature Engineering, Streamlit GitHub

- –Developed an end-to-end Machine Learning system to predict **hospital Length of Stay (LOS) categories** and **patient billing amounts** using classification and regression models trained on healthcare records.
- –Built a data preprocessing and feature engineering pipeline using **Pandas and Scikit-learn**, handling missing values, categorical variables, and extracting temporal features including admission month, quarter, and day-of-week.
- –Implemented a **chained prediction architecture** where the predicted LOS category is used as an additional feature for the billing regression model to capture dependencies between hospital stay duration and billing.
- –Achieved **85% classification accuracy** for LOS prediction and **R^2 of 0.87** for billing prediction; deployed the complete inference pipeline through an interactive **Streamlit application**.

Education

Degree/Certificate	Institute / Board	CGPA / Percentage	Year
B.Tech – CSE	Indian Institute of Information Technology, Una	**8.88 / 10.0**	2024–Present
Senior Secondary (XII)	CBSE Board	**91.4%**	2024

Achievements & Activities

•**250+ DSA Problems**, Solved using C++, covering Graphs, Trees, DP,Binary Search, Greedy Algorithms *2025–26*
•**Kaggle Clash** , Designed and conducted a Kaggle-based Machine Learning competition for 30+ participants *2026*
•**Robo Trace**, Secured **1st place** in the Line Follower Robot competition at Meraki, the college technical festival *2026*
•**Run Code Run**, Organised a competitive coding and algorithmic problem-solving event for 40+ participants *2025*